

Mathematics And Physics For Direct Entry Notes

Angus

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Abstract

This is a simple abstract for the Mathematics And Physics For Direct Entry notes.

This note is for physics part.

1 Lecture 1

2 Lecture 2

2.1 Fourier series

Combination of harmonics to represent a function. components have different amplitudes, need to be correct amplitudes.

2.2 The Fourier principles

Work in practice and do not work in theory.

Any wave front can be represented as a superposition of harmonics(different harmonic frequencies(or interchangeably, wavelengths) sin waves).

So for example, triangular wavefront $y=ax$ form $[0, L]$ and zero at another region.

$$y(x) = \sum_{n=1}^{\infty} y_n \sin\left(\frac{n\pi x}{L}\right)$$

y_n is the amplitude of the nth harmonic component. $n = 1, 2, 3, \dots$ so n can not be zero. maximum value can not be reached because we can not reach infinite harmonics.

$$\int y(x) \sin\left(\frac{m\pi x}{L}\right) dx = \sum y_n \int \sin\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi x}{L}\right) dx$$

use

$$\sin\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi x}{L}\right) = \frac{1}{2} \left[\cos\left(\frac{(m-n)\pi x}{L}\right) - \cos\left(\frac{(m+n)\pi x}{L}\right) \right]$$

and integrating right hand side.

$$\begin{aligned} \int \sin\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi x}{L}\right) dx &= \frac{L}{2\pi(n-m)} \sin\left(\frac{(n-m)\pi x}{L}\right) \\ &\quad - \frac{L}{2\pi(n+m)} \sin\left(\frac{(n+m)\pi x}{L}\right) \end{aligned}$$

But if $n=m$ the equation is undefined. So we need to consider only the $n=m$ harmonic term. So if we now consider just the $n=m$ harmonic term and ignore other harmonic terms in the sum/superposition (as they will contribute zero to the integral), we now solve the integral for $n=m$.

$$\int \sin^2\left(\frac{n\pi x}{L}\right) dx = \frac{1}{2} \int (1 - \cos\left(\frac{2n\pi x}{L}\right)) dx = \frac{L}{2}$$

Inserting the above result in equation below

$$\int y(x) \sin\left(\frac{m\pi x}{L}\right) dx = \sum y_n \int \sin\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi x}{L}\right) dx$$

We can then show that for a generalised waveform, $y(x)$, bounded within the limits $x = 0 \rightarrow L$, the amplitude of the n th harmonic is given by

$$y_n = \frac{2}{L} \int y(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

So for the example of a triangular wavefront, we can calculate the amplitudes of its harmonic components using the formula above, and use integration by parts to evaluate the integrals.

$$\begin{aligned} y_n &= \frac{2}{L} \int ax \sin\left(\frac{n\pi x}{L}\right) dx \\ y_n &= \frac{2a}{n\pi} \left[-x \cos\left(\frac{n\pi x}{L}\right) + L/n\pi \sin\left(\frac{n\pi x}{L}\right) \right] \Big|_0^L \\ y_n &= \frac{-2aL}{n\pi} \cos(n\pi) \\ y(x) &= \sum_{n=1}^{\infty} \frac{-2aL}{n\pi} \cos(n\pi) \sin\left(\frac{n\pi x}{L}\right) \end{aligned}$$

And in mechanic system each harmonic terms so-called normal modes of vibration.

2.3 Casimir effect

Two metal plates placed close to each other in a vacuum experience an attractive force.

For reasons, outside the plates, more states are possible, but between the plates, there is only a certain discrete set of states allowed. This difference in the density of states leads to a net attractive force between the plates.

There is nothing in vacuum.

2.4 Classical wave theory of light

Light is treated as a wave, and phenomena such as interference and diffraction are explained using wave principles.

2.4.1 Huygens principle

Huygens' principle states that every point on a wavefront can be considered as a source of secondary spherical wavelets, and the new wavefront is the envelope of these wavelets. the wavefront represents a surface/region of space in which the wave has the same instantaneous phase.

3 Lecture 3

3.1 One slit diffraction of Light

Single narrow slit diffraction occurs when light passes through a slit whose width is comparable to the wavelength of the light, resulting in a diffraction pattern on a screen placed behind the slit.

$$\text{the Optical path difference} = \frac{a}{2} \sin \theta$$

produces a phase difference.

Destructive interference occurs when the path difference is an odd multiple of half the wavelength, leading to dark fringes on the screen.

$$a \sin \theta = \lambda$$

Constructive interference occurs when the path difference is an integer multiple of the wavelength, leading to bright fringes on the screen.

3.1.1 Diffraction of Photon

For a photon passing through the slit, the diffraction pattern can be understood as the result of the interference of the wavefronts emanating from different parts of the slit. perspective of particle

$$\Delta x \sim a$$

for a photon

$$E = pc$$

Direction of photon can be different.

De Broglie relation,

$$\Delta p_x \sim \frac{h}{\lambda} \sin \theta$$

so wider slit leads to greater uncertainty in the photon's momentum direction.

for $a \sin \theta = \lambda$

$$\Delta x \Delta p_x \sim h$$

we can use classical mechanism in every scale when the plank constant equals to zero.

3.1.2 Fresnel Bright Dot

Fresnel bright dot, also known as the Poisson spot, is a bright point that appears at the center of the shadow of a circular obstacle due to the constructive interference of diffracted light waves.

Is a direct consequence of Huygens' principle and the wave nature of light.

Light does not always travel in straight lines.